



Indian Institute of Foreign Trade, Kolkata

A Machine Learning Framework for Credit Risk Prediction Integrating Financial and Transactional Behavior



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Transaction-Augmented Credit Risk: Hypothesis & Business Case

Moving beyond static snapshots to capture real-time behavioral volatility.

Strategic Context

Traditional credit models rely on static application data (demographics, loan history), creating blind spots for borrowers with deteriorating real-time cash flow or “bust-out” trajectories (**Coral: Risk/Default**).

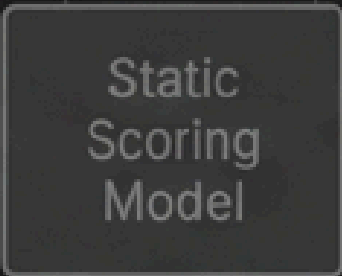
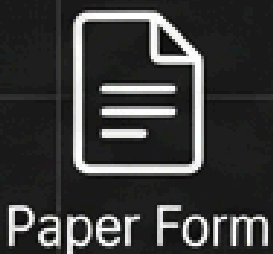
Objective

Test the hypothesis that high-frequency transaction data (volatility, velocity, volume) provides orthogonal predictive power over standard variables.

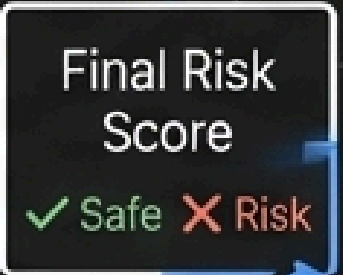
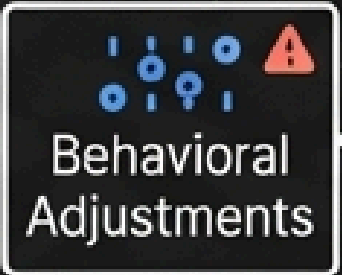
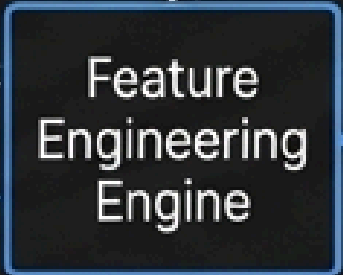
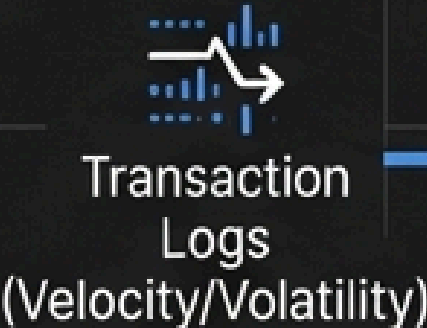
Key Drivers

- 1. **Hidden Risk Detection (Coral):** Identifying borrowers with prime paper scores but high behavioral volatility.
- 2. **Thin-File Assessment (Steel Blue):** Augmenting decisioning for applicants with limited credit history.
- 3. **Affordability vs. Behavior:** Distinguishing between structural capacity to pay (**Sage Green**) and behavioral willingness to pay.

Traditional Assessment



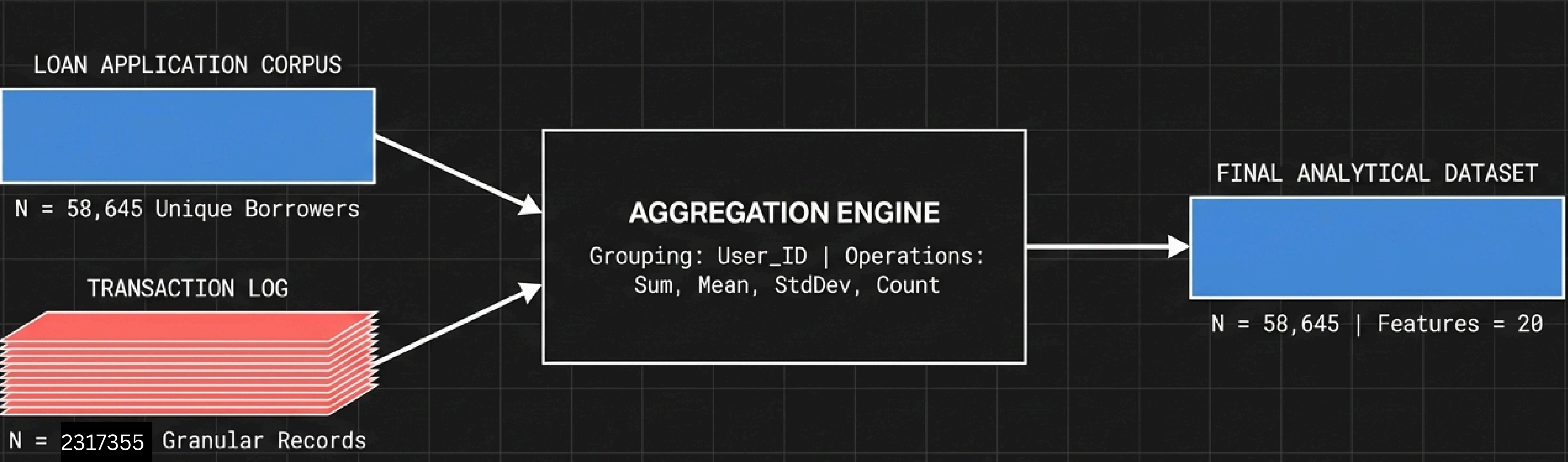
Augmented Assessment



Hypothesis Test: **+Alpha?**

DATA ARCHITECTURE: MERGING STATIC PROFILES WITH BEHAVIORAL LOGS

Engineering a unified view from disparate data granularities.



METRIC	VALUE	ENGINEERING NOTE
Observations	58,645 Borrowers	Clean application data, no missing criticals.
Transaction Volume	2317355 Events	High-frequency logs converted to borrower-level signals.
Merge Strategy	One-to-Many Aggregation	Flattened using statistical moments (Mean/Volatility).
Final Dimensionality	20 Features (11 Static + 9 Behavioral)	Merged on unique User_ID key.

FEATURE COMPOSITION & TARGET VARIABLE DISTRIBUTION

Defining the modeling inputs and the class imbalance challenge.

FEATURE TAXONOMY

STRUCTURAL / STATIC (APPLICATION DATA)

- Loan Amount, Interest Rate, Annual Income
- Employment Length, Home Ownership Status
- Loan Grade, Historical Defaults

BEHAVIORAL / DYNAMIC (DERIVED TRANSACTION DATA)

- Total Spend Volume
- Transaction Volatility (Std Dev of Amount)
- Transaction Count / Intensity
- Geographic Diversity

NOTE: Outliers in `spend\_volatility` and `loan\_to\_income` explicitly retained as economic stress signals.

TARGET VARIABLE: LOAN_STATUS

86%

14%

CLASS 0: NON-DEFAULT

Majority Class

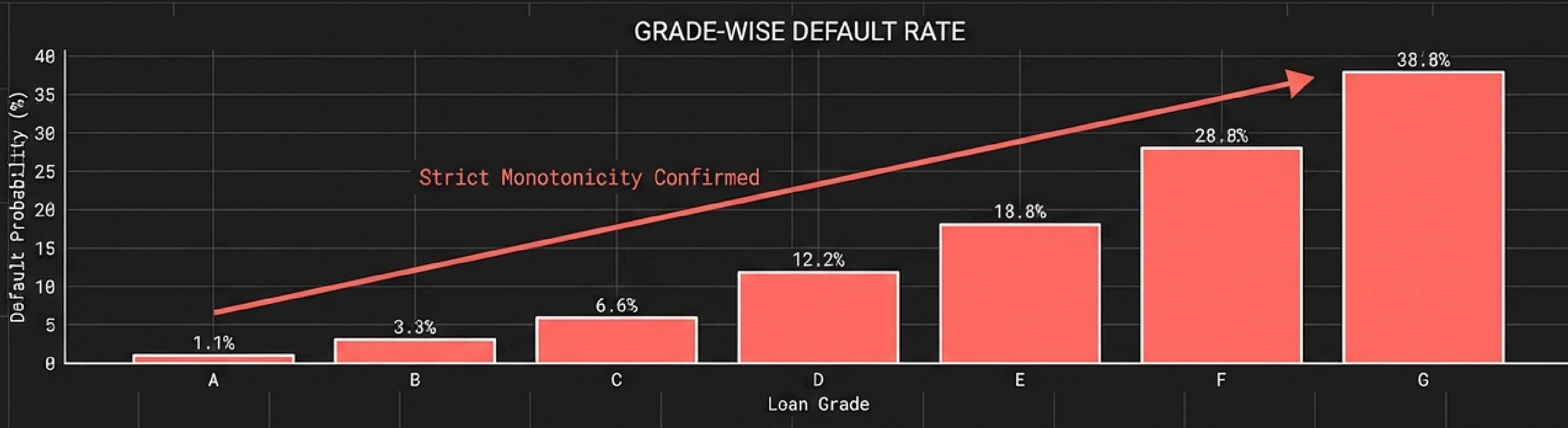
CLASS 1: DEFAULT

Target Minority Class

MODELING IMPLICATION: 14% Default Rate necessitates Stratified Sampling and Class Weight adjustments in Random Forest.

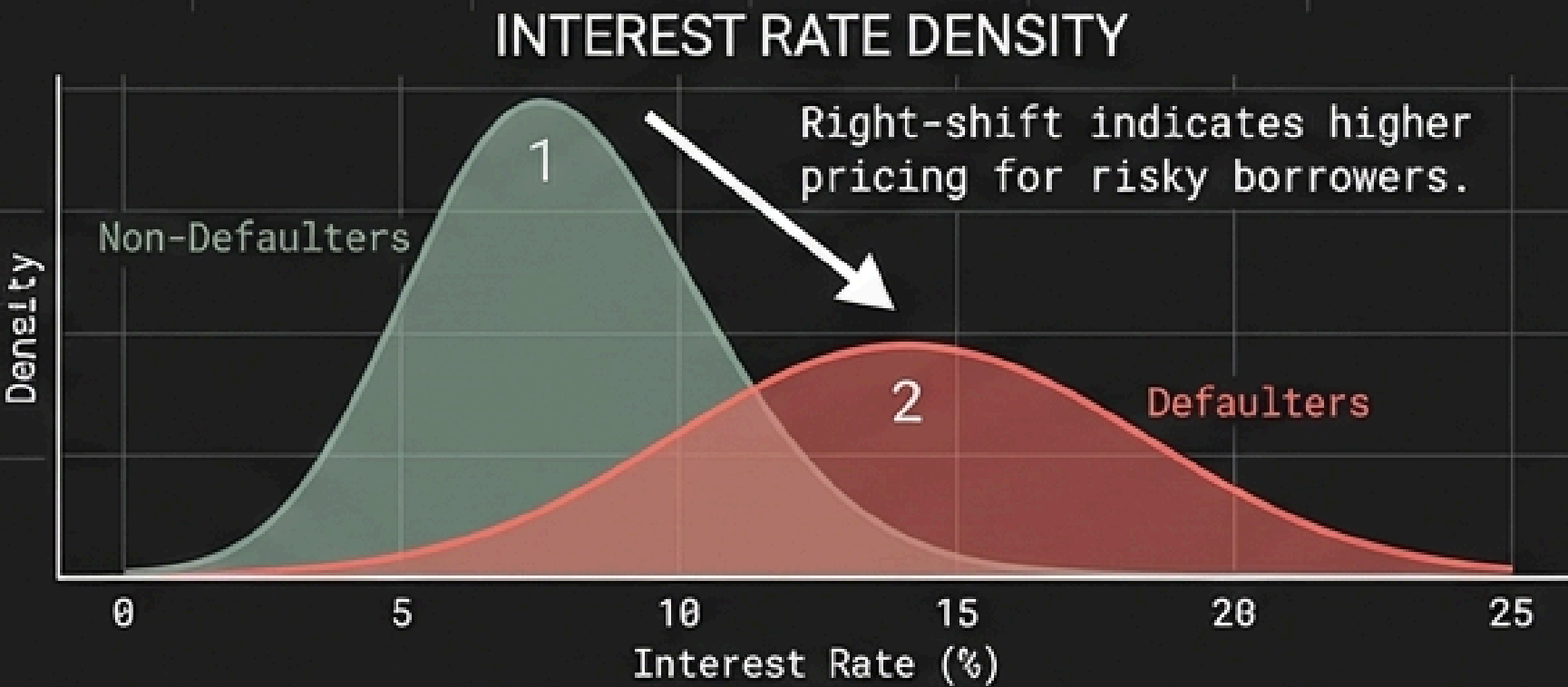
EXPLORATORY ANALYSIS: VALIDATING TRADITIONAL RISK SIGNALS

Confirming monotonicity in Grades and pricing mechanisms.



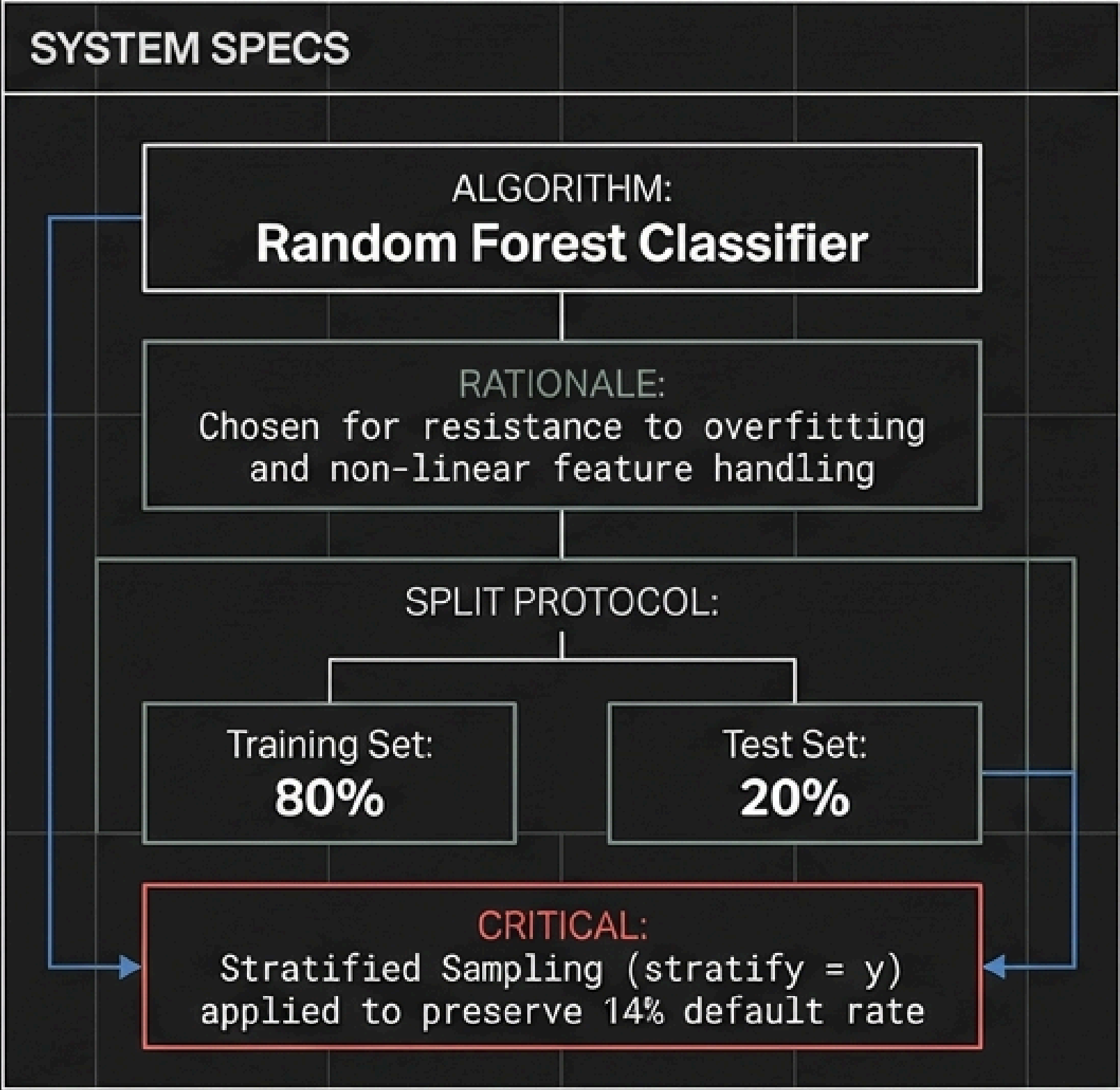
KEY OBSERVATIONS

- **Affordability Stress:** Defaulters exhibit significantly higher Loan-to-Income (LTI) ratios with wider dispersion.
- **Pricing Correlation:** Default density is concentrated in higher interest rate bands, validating existing risk-based pricing.
- **Conclusion:** The dataset accurately reflects fundamental economic logic.



BASELINE MODEL SPECIFICATION (STATIC DATA ONLY)

Establishing a rigorous control model using Random Forest



CONFIGURATION PARAMETERS		
Parameter	Value	Impact
n_estimators	100 Trees	Ensures ensemble stability.
class_weight	"Balanced"	Penalizes misclassification of minority class (Defaulters).
random_state	42	Reproducibility.
criterion	Gini Impurity	Split quality measurement.

BASELINE PERFORMANCE: STRONG GLOBAL METRICS, SPECIFIC BLIND SPOTS

The model is robust (AUC 0.93) but suffers from a critical precision gap.

Performance Scorecard

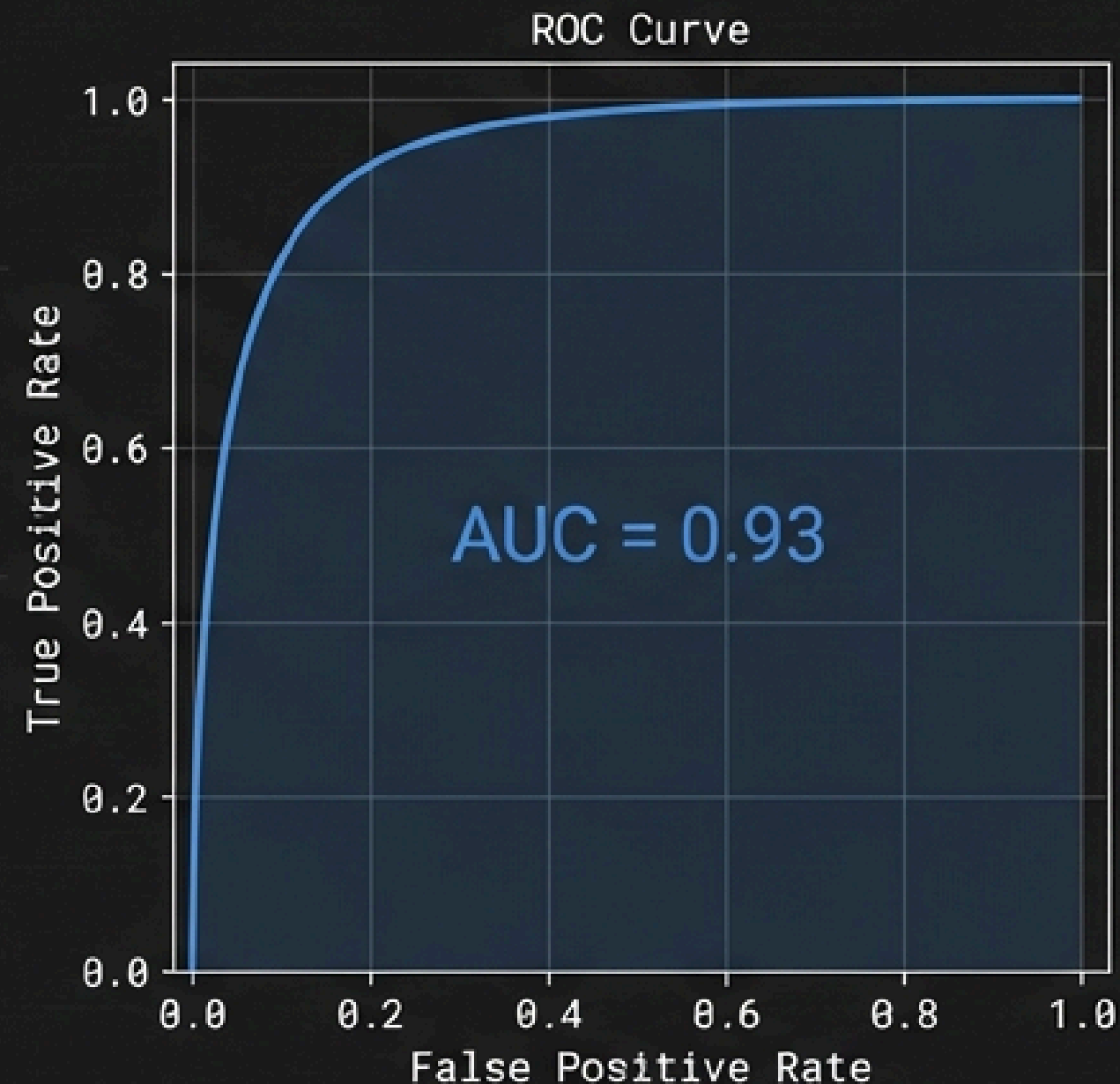
ROC-AUC: 0.9323

Strong Separation.

Accuracy: 95.11%

F1 Score: 0.8052

The "Precision Gap" Analysis



The Blind Spot:
Precision = 0.71

While global accuracy is high, the model **fails to identify 29%** of actual defaulters. These are who appear "Safe" on paper (Application Data) but carry hidden risk.

This gap motivates the transaction data augmentation.

THE GLOBAL PREDICTION LIMIT: WHY MORE DATA ≠ BETTER ACCURACY

The enhanced model failed to outperform the baseline due to signal redundancy.



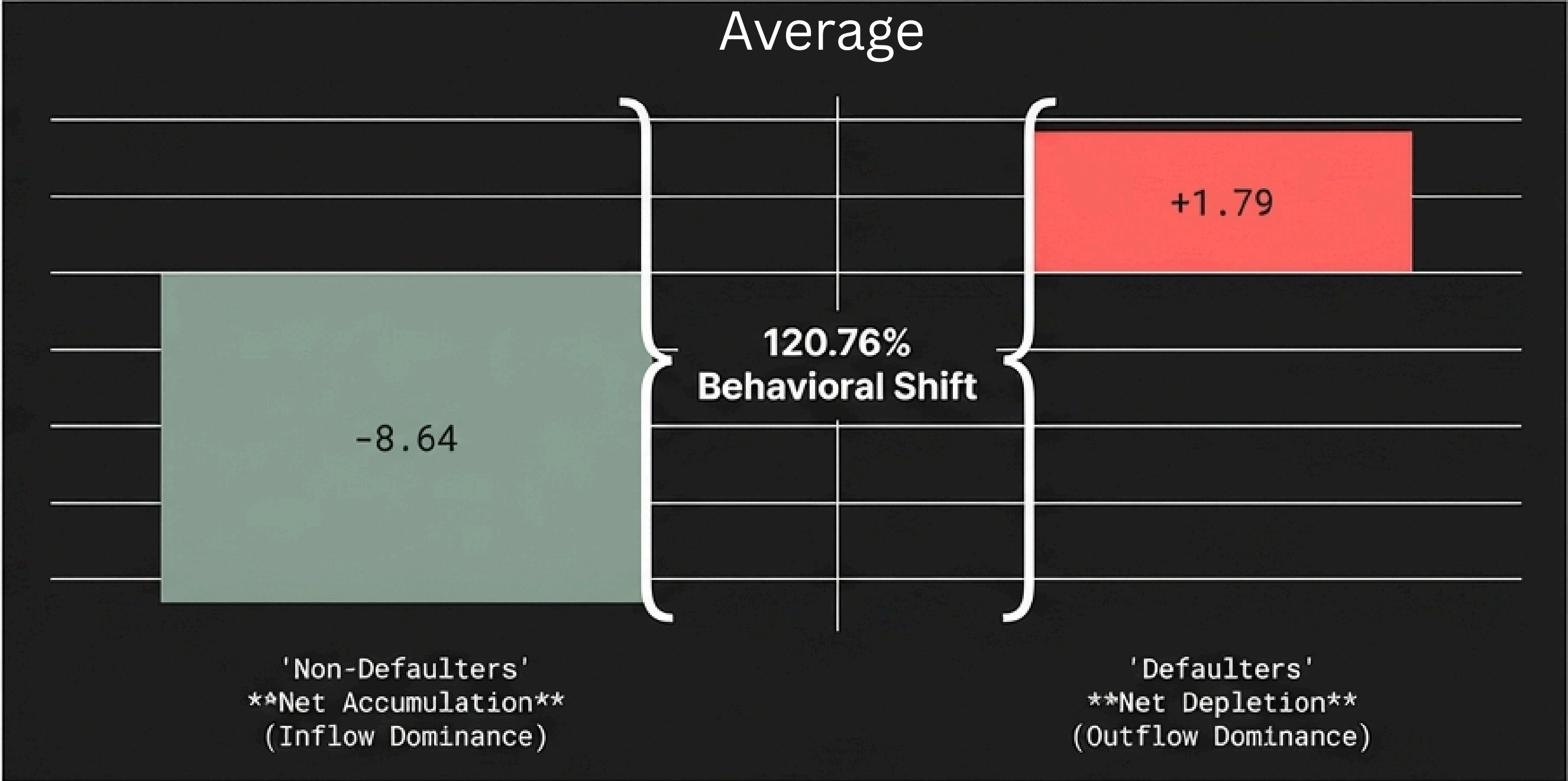
DIAGNOSTIC ANALYSIS		
REDUNDANCY High correlation found between `Total_Spend` and `Person_Income`. Transaction volume acted as a proxy for income rather than a new signal.	SPARSITY NOISE Thin transaction histories for subsets of users required zero-filling/averaging, introducing noise that diluted strong static signals.	BASELINE STRENGTH The static model (AUC 0.93) was already capturing the primary default drivers (Affordability), leaving little room for marginal improvement via volume metrics.

BEHAVIORAL ALPHA I: THE TRANSACTION VOLATILITY COEFFICIENT (TVC)

Revealing a 120% polarity shift in financial management style.

$$\text{TVC} = \frac{\text{Spend Volatility } \sigma}{\text{Average Transaction Amount } \mu}$$

Definition: Measures the directional stability of cash flow.



High risk borrowers do not simply spend more or less. They just operate with a completely different financial direction. Safe borrowers tend to have net inflows while risky borrowers experience net outflows.

Risk is not just about spending *amount*; it is a fundamental directional difference. Safe borrowers accumulate liquidity (Negative TVC); risky borrowers hemorrhage liquidity (Positive TVC). This signal is orthogonal to income.

BEHAVIORAL ALPHA II: INTRADAY SPENDING VELOCITY (ISV)

Identifying "Bust-Out" risk through temporal acceleration patterns.

THE METRIC

ISV Definition:

Percentage drift in mean transaction amount from Hour 0 to Hour 14.

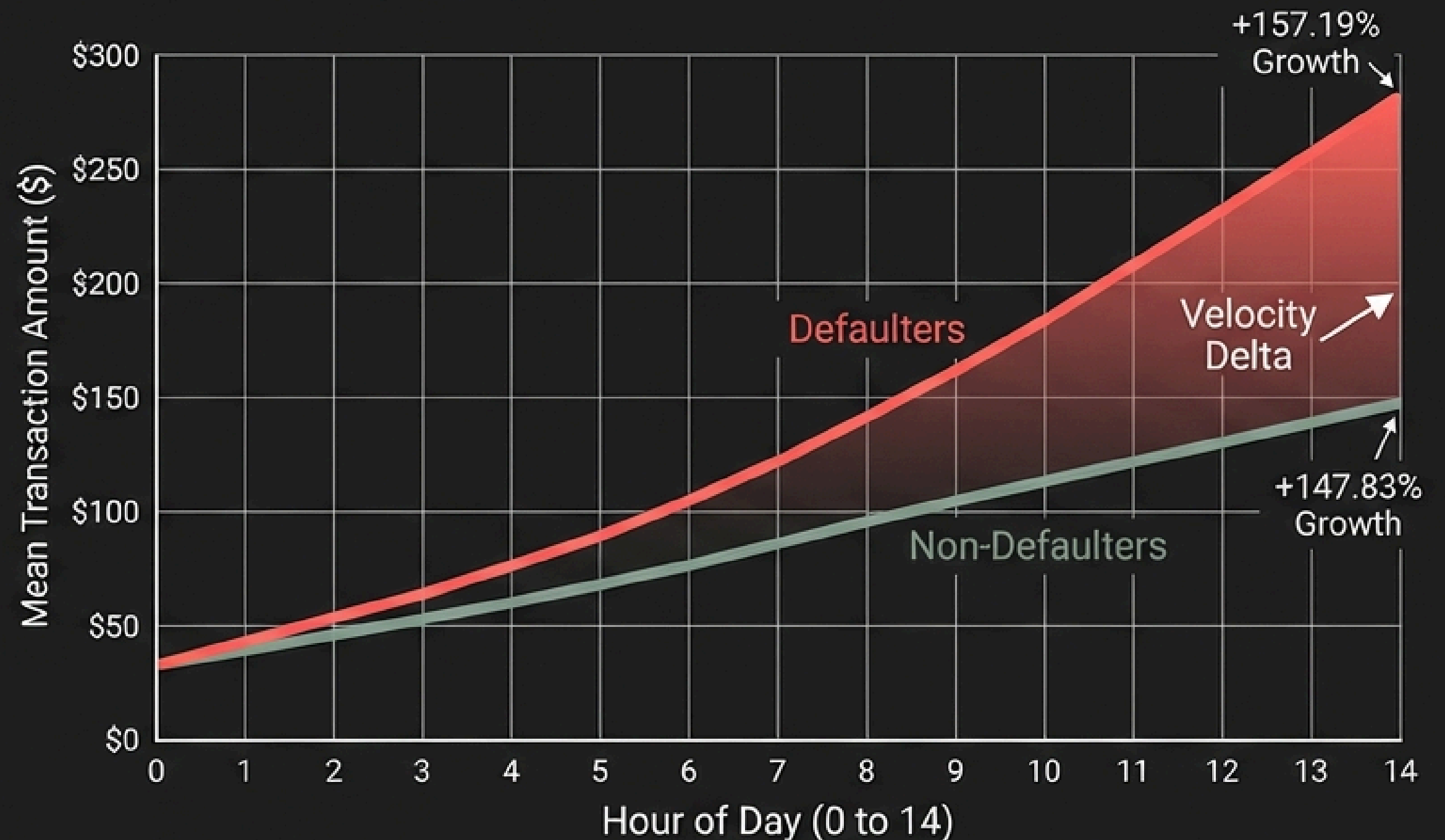
The Signal:

Defaulters accelerate spending **9.36%** faster than non-defaulters as the day progresses.

Interpretation:

Consistent with "bust-out" fraud or impulsive liquidity draining.

TEMPORAL DIVERGENCE



STRUCTURAL DOMINANCE: THE HOME OWNERSHIP RISK MULTIPLIER

Asset backing overrides behavioral nuance as the primary risk determinant.

RENTERS

Population: N = 30,594

22.26%

Avg Volatility: 77.81

OWNERS

Population: N = 3,138

1.37%

Avg Volatility: 80.03

16.2x

1,524% Higher Risk for Renters

Crucial Finding: Spending Volatility difference is negligible (-2.8%).
Risk is driven by **Wealth Structure**, not Daily Behavior.

The Home Ownership Multiplier

	Segment A(Renters)	Segment B(Owners)
Count	30,594	3138
Default Rate	22.60%	1.37%
Avg. Volatility	77.81	80.03
Risk	High	Low

- The analysis reveals a 1,524 divergence in risk outcomes in spite of nearly identical spending behavior
- Talking about the sensitivity, the renters are 16.2x more likely to default than owners (a + 1,524.17% risk multiplier).
- The analysis demonstrates the extreme outcomesensitivity to asset ownership. The renters and owners exhibit a -2.8% difference in daily spending variance (specifying similar financial behavior), where the risk profiles of both are radically different. This proves that person_home_ownership is a critical discriminator

Model Robustness & Stability Analysis

Stability Across Data Splits (5-Fold Stratified Cross-Validation)

Results:

- Mean ROC-AUC:0.9247
- Standard Deviation: 0.0042 ($\pm 0.42\%$)

1. The model shows high stability across different training-validation splits, with a mean ROC-AUC of 0.9247 and very low variance ($\pm 0.42\%$).
2. This indicates that performance is consistent and not dependent on a specific data partition, confirming reliable generalization.

Robustness to Noisy Inputs (Stress Test)

Results:

- Mean ROC-AUC:0.9247
- Standard Deviation: 0.0042 ($\pm 0.42\%$)

1. After injecting 20% Gaussian noise into the key income feature, the model's ROC-AUC declined only marginally from 0.9247 to 0.9177.
2. The $<1\%$ performance drop shows that the model retains over 99% of its predictive power, even under degraded data conditions.

The model is stable, robust, and well-generalized, with decision boundaries that are resilient to both data variability and feature noise, making it suitable for real-world deployment.

INTERACTION ANALYSIS: HOUSING STATUS AS A RISK MODERATOR

DTI > 0.30 is fatal for Renters but manageable for Owners.

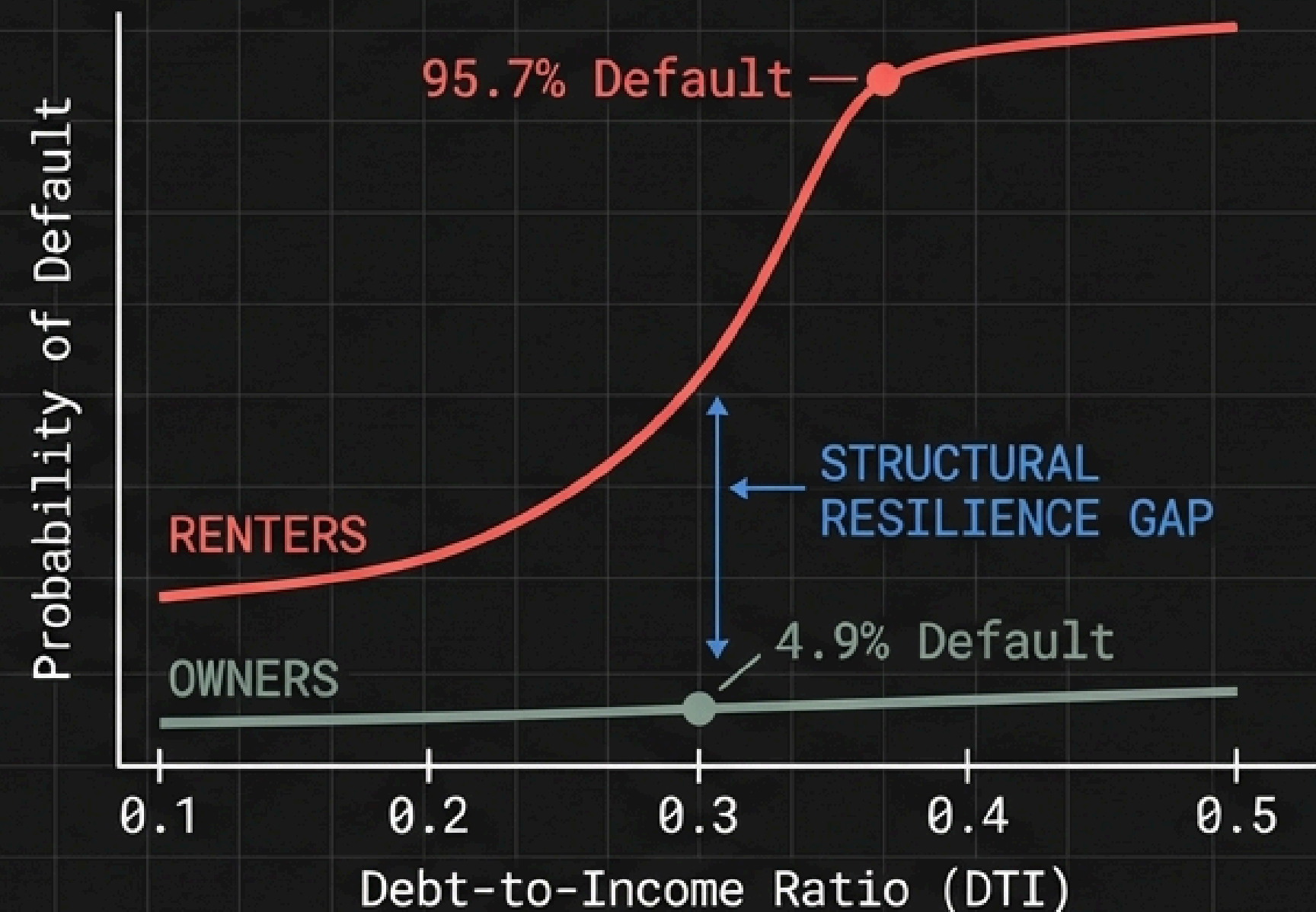
LOGISTIC REGRESSION
ADJUSTED ODDS RATIO
(RENT VS OWN):
30.49x ($p < 0.001$).

THE MISCLASSIFICATION TABLE:
FAILURE OF CURRENT GRADING

GRADE A RENTERS	ACTUAL DEFAULT: 10.66%
GRADE B OWNERS	ACTUAL DEFAULT: 0.56%

Insight: Prime-Grade Renters are 18.9x riskier than Sub-Prime Owners.

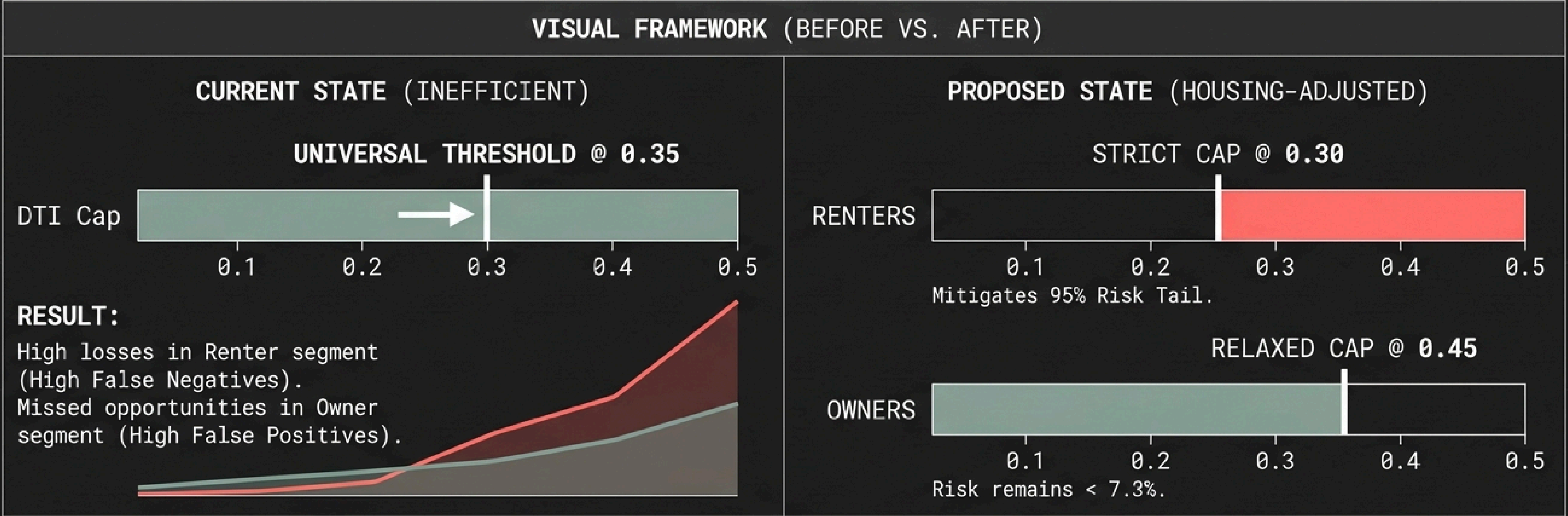
INTERACTION EFFECT VISUALIZATION



STRATEGIC POLICY RECOMMENDATIONS: A TWO-TIERED RISK FRAMEWORK

Operationalizing structural insights to optimize portfolio performance

Core Conclusion: Financial Structure (Assets) > Behavioral Nuance (Transactions).
While Transaction data adds color, Housing Status must anchor the credit policy.



PROJECTED IMPACT
Elimination of high-probability losses (High-DTI Renters) and expansion of loan book into low-risk, high-DTI Homeowners.

Conclusion: Financial Structure vs. Behavioral Nuance

1. Traditional Metrics Remain the "Gold Standard"

- **Predictive Power:** The baseline model using only 11 application variables (e.g., Loan Grade, Debt-to-Income) achieved a strong ROC-AUC of 0.93.
- **Marginal Gains from Transactions:** Incorporating transaction data (totaling 20 features) did not improve accuracy (95.11% vs 94.94%). High correlations between spending and income rendered transaction data redundant for prediction.

2. Behavioral Insights Revealed "Polarity Shifts"

- While not increasing predictive accuracy, behavioral data exposed how borrowers manage money differently.
- **Transaction Volatility Coefficient (TVC):** Defaulters show a positive TVC (spending spikes with consumption), while non-defaulters show a negative TVC (inflows dominate), marking a fundamental 120% behavioral shift.
- **Temporal Risk:** Defaulters exhibit a "Bust-Out" dynamic, accelerating spending 9.36% faster than non-defaulters as the day progresses.

3. Housing Status is a Critical Risk Moderator

- **The Renter Risk Gap:** Renters are approximately 30.5 times more likely to default than homeowners when controlling for income and age.
- **Grade Misclassification:** Current scoring underestimates renter risk; "Grade A" renters default at 10.66%, while "Grade B" owners default at only 0.56%.

Strategic Recommendation

Transition to Housing-Adjusted DTI Thresholds Instead of a universal Debt-to-Income (DTI) cap, the credit policy should be bifurcated based on asset ownership:

- **For Renters:** Implement a strict DTI cap of 0.30. (Risk exceeds 95% beyond this threshold).
- **For Homeowners:** Allow a relaxed DTI cap of 0.45. (Risk remains below 7.3% even at higher leverage).

Thank You !!!